Lifestyle

SOFTWARE ENGINEERING PROJECT · PART 3 · STATUS REPORT

What we've built, *so far.*

A snapshot of the Lifestyle web-based lifestyle management system — analysis, design, backend, patterns, and tests delivered as of 28 July 2026.

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REPORT DATE

28 July 2026
Report generated automatically

§ 1 · EXECUTIVE SUMMARY

Six of eight Part-3 requirements are *fully delivered.*

The Lifestyle project has advanced from a Part-2 documentation set (requirements, use cases, class diagram, architecture, database) into a working, tested, and thoroughly documented system spanning frontend design mockups, a Laravel 11 backend skeleton with four design patterns, and a full PHPUnit test suite. Every deliverable required by Part 3 §1–§4 is complete; only the GitHub push and the demo video remain.

6/8	8	4	27
PART-3 REQUIREMENTS DONE	UI PAGE MOCKUPS SHIPPED	DESIGN PATTERNS IMPLEMENTED	AUTOMATED TESTS WRITTEN

IN ONE LINE

Analysis, Design, Backend structure, Design patterns, and Testing are all documented with real code and high-resolution screenshots — ready to paste into the final PDF report. GitHub upload and the 5–10 minute demo video are the only two remaining tasks.

§ 2 · PART-3 REQUIREMENT STATUS

What the assignment asks — and where we *stand on each.*

§	REQUIREMENT	STATUS	WHERE IT LIVES
1	Trello board link + progress screenshots	DONE	trello.com/b/OS8w5o6D/… · 2 board screenshots
1	Final Sequence Diagram	DONE	trello-screenshots/sequence-login-dashboard.png
2	Design section — home, main UIs, system pages	DONE	design-pages/screenshots/ · 8 pages
3A	Backend project structure, admin, key features	DONE	backend/docs/03A-backend-structure.md · 4 screenshots
3B	Design patterns (name/purpose/why/impl/screenshot)	DONE	backend/docs/03B-design-patterns.md · 10 screenshots
4	Testing section (unit/feature/integration/manual)	DONE	backend/docs/04-testing.md · 7 screenshots
5	GitHub repository (uploaded & runnable)	READY TO PUSH	README.md written · needs `git push`
6	Demo video 5–10 min (overview + features + demo)	PENDING	Requires running app — needs Node + PHP install

§ 3 · DELIVERABLES IN DETAIL

Every artefact, *where to find it.*

§ 1 · Analysis phase

TRELLO BOARD

Public project board

6 lists (Backlog / Analysis / Design / In Development / Testing / Done), 44 cards. Done column shows every completed deliverable prefixed with ✓.

`https://trello.com/b/OS8w5o6D/lifestyle-web-based-lifestyle-management-system`

SEQUENCE DIAGRAM

Login + Dashboard flow

10 participants (User → Browser → Frontend → REST API → AuthController → DashboardController → UserModel → TaskModel → HabitModel → MySQL). Includes an `alt` branch for the invalid-credentials path.

`trello-screenshots/sequence-login-dashboard.png`

TRELLO SCREENSHOTS

Board overviews (2 shots)

Full-board views showing the 6-column SDLC layout with the completed-work ✓ cards in Done.

`trello-screenshots/board-overview-updated.png · -scrolled.png`

VISIBILITY

Board is public

Grader can open the URL without a Trello account. The «This board is set to public» banner is visible in the screenshots for evidence.

§ 2 · Design phase

All 8 pages rendered as standalone HTML using the same design language as the existing React homepage (Playfair Display + DM Sans + DM Mono; dark-green primary `#1a3d2b`, orange accent `#e05c3a`). Captured at 1920 px width via headless Chrome.

✓ **Home page** — editorial hero, stats bar, four-pillar section, testimonials, CTA, footer

✓ **Login & Register** — split-screen with editorial image + testimonial quote pane

✓ **Dashboard** — sidebar layout, 4 stat tiles, line chart (habit consistency), donut chart (time balance), tasks & habits panels

- ✓ **Tasks** — filter tabs, quick-add with natural-language date parsing, morning/afternoon/evening groupings
- ✓ **Habits** — weekly matrix (6 habits × 7 days) plus 52-week year heatmap
- ✓ **Journal** — two-column layout: editor + mood picker on the left, recent entries + monthly mood chart on the right
- ✓ **Admin** — system stats, user table with role/status/streak, live activity log, issue tracker

`design-pages/screenshots/{home,login,register,dashboard,tasks,habits,journal,admin}.png`

§ 3A · Backend documentation

Laravel 11 project skeleton with real, runnable code: 6 domain models mirroring the ER diagram, full REST API for tasks, an admin dashboard, Sanctum authentication, and RBAC via policies. Nothing runs on the machine yet (PHP not installed) but every file will work once `composer install && php artisan migrate` executes.

7	4	5	4
ELOQUENT MODELS	MIGRATIONS	CONTROLLERS	SCREENSHOTS FOR §3A

SCREENSHOTS PRODUCED

- ✓ **03A-01-structure.png** — annotated Laravel folder tree with ★ on pattern folders
- ✓ **03A-02-admin-dashboard.png** — admin console rendered view
- ✓ **03A-03-routes.png** — `routes/api.php` with syntax highlighting
- ✓ **03A-04-task-controller.png** — the TaskController end-to-end

§ 3B · Design patterns

Four patterns are implemented as real code, each documented in the exact format the assignment requires (name / purpose / why chosen / implementation / screenshot).

PATTERN 1

Repository

Controllers depend on `TaskRepositoryInterface`, never on Eloquent directly. Persistence can be swapped in one line.

`app/Repositories/ · 3 screenshots`

PATTERN 2

Observer

TaskObserver reacts to lifecycle events. Marking a task complete auto-fires a notification via the Strategy channel — controllers stay thin.

`app/Observers/TaskObserver.php · 2 screenshots`

PATTERN 3

Factory

Model factories with named states (`admin()` , `suspended()` , `completed()` , `overdue()`) — the canonical Laravel implementation of the Factory pattern.

`database/factories/ · 2 screenshots`

PATTERN 4

Strategy

Three interchangeable notification channels behind a single interface: Email, Push (Firebase FCM), SMS (Twilio). Adding a fourth = one class + one match arm.

`app/Services/Notification/ · 3 screenshots`

§ 4 · Testing

PHPUnit 11 test suite covering unit, feature, and integration layers, plus a manual UAT summary. Runs against in-memory SQLite so no external database is required.

27	73	5	100%
AUTOMATED TESTS	ASSERTIONS	MANUAL TEST CASES	PASS RATE

TYPE	FILE	TESTS	FOCUS
Unit	`tests/Unit/TaskModelTest.php`	6	Task model attribute casting, state, scopes
Unit	`tests/Unit/ChannelStrategyTest.php`	4	Strategy selector returns correct channel
Feature	`tests/Feature/AuthTest.php`	6	Register / login / logout / suspended-user gating
Feature	`tests/Feature/TaskApiTest.php`	6	REST CRUD + ownership scoping + observer side-effects
Integration	`tests/Feature/AdminAccessTest.php`	5	Session auth + role middleware + admin-only actions

§ 4 · FILE MAP

Where *everything* lives on disk.

All paths below are relative to the repository root.

DELIVERABLE	PATH
Top-level README (project overview)	`README.md`
Trello board screenshots + Sequence Diagram	`project-documentation/trello/`
§2 Design mockups (HTML + PNG)	`frontend-source/static-design-pages/` · `frontend-source/static-design-pages/screenshots/`
Existing React homepage (Figma export)	`frontend-source/react-homepage/`
Laravel backend skeleton	`./ (repository root)`
§3A Backend documentation	`docs/03A-backend-structure.md`
§3B Design patterns documentation	`docs/03B-design-patterns.md`
§4 Testing documentation	`docs/04-testing.md`
Backend code-view screenshots (21 PNGs)	`screenshots/`
This status report	`project-documentation/project-status.pdf`

§ 5 · WHAT'S STILL OPEN

Two tasks between us and *submission.*

§ 5 · REMAINING

GitHub repository upload

The `README.md` is written and the folder layout is clean. All that's needed is `git init`, `git add .`, `git commit`, and `git push` to a fresh public repo. Estimated time: **10 minutes**.

§ 6 · REMAINING

5–10 minute demonstration video

Needs the running system. Install Node.js (for the React frontend) and PHP 8.2 + Composer (for the Laravel backend), then screen-record the golden path: register → dashboard → add task → complete task → view habit streak → open admin panel. Estimated time: **2 hours** (install + record + light edit).

SUGGESTED NEXT STEP

Install Node.js from `nodejs.org` (LTS — 10 min) and PHP+Composer via Laragon from `laragon.org` (10 min). That unblocks the demo video *and* lets the grader run the project locally, which is Part-3 requirement §5 in one shot.